

A Boundary-Style Interpretation of Quantum Measurement

A Gentle Guide to Sampling, Boundary Access, and the Appearance of Randomness

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This note offers a conceptual lens on how quantum measurement can be understood through the same interior–boundary asymmetry that appears in Gauss’s law. The goal is not to propose a new ontology or a competing interpretation, but simply to show a structural pattern that readers may find clarifying.

In Gauss’s law, the observer never inspects the interior charge distribution. Only the outward-facing flux at the boundary is accessible; the interior is inferred. This asymmetry—*boundary accessible, interior hidden*—turns out to be a surprisingly helpful way to think about quantum measurement as well. The wavefunction becomes a boundary signature, measurement becomes a boundary comparison, and randomness becomes a feature of how the observer samples the boundary rather than a statement about the interior.

1. The Boundary as the Only Accessible Interface

Imagine a region whose interior evolves in ways we cannot see. We do not track the microscopic details; we only observe what reaches the boundary.

This is exactly how Gauss’s law works. The observer measures the outward flux, not the interior charge distribution. The mapping from interior to boundary is many-to-one, and that is the whole point.

Quantum systems can be viewed the same way. The wavefunction is not a map of the interior. It is the **outward-facing signature**—the pattern of information that reaches the boundary and becomes available to an observer.

Many different interior configurations may produce the same boundary signature. This is not a flaw. It is the same structural asymmetry that Gauss’s law expresses.

2. A Continuously Evolving Boundary Signature

Even the simplest quantum system produces a boundary signature that shifts over time.

For a two-level system,

$$|\psi(t)\rangle = ae^{-i\omega_1 t}|1\rangle + be^{-i\omega_2 t}|2\rangle,$$

the relative phase

$$\Delta\phi(t) = (\omega_2 - \omega_1)t$$

changes continuously.

From the boundary’s perspective, the outward-facing pattern is always in motion.

The interior may evolve smoothly, but the observer only ever sees its changing boundary imprint.

This is the first key idea:

the boundary is dynamic, even when the observer is not watching.

3. What a Measurement Really Samples

In Gauss’s law, a measurement is not a pointlike event.

It is an **integral over a surface**—a gathering of boundary information.

Quantum measurement can be understood in the same spirit.

A detector does not take an instantaneous snapshot.

It interacts with a finite patch of the boundary over a finite duration.

What we call a “measurement outcome” is the result of this **sampling process**.

Let P be the spacetime patch where the detector couples to the system.

The detector effectively accumulates a boundary signal:

$$\mathcal{M}(\psi) \sim \iint_P F(B\psi(x, t)) dA dt.$$

This is not a new physical law.

It is simply the Gauss-style idea that measurement is a **boundary-level integration**, not a direct probe of the interior.

4. Why Randomness Appears — A Boundary-First View

Here is the heart of the matter.

Because the boundary signature evolves continuously, and because the detector samples only a finite patch of it, the exact outcome depends on the **micro-timing** and **micro-geometry** of the sampling window.

The observer does not track the boundary continuously.
They sample it coarsely, asynchronously, and with limited resolution.
The resulting variability appears as randomness.

From this perspective:

Quantum randomness is not a mysterious property of nature.

It is the natural consequence of sampling a continuously evolving boundary with finite resolution.

This is not a hidden-variable claim.

It is simply the Gauss structure applied to measurement:

the observer never sees the interior, and the boundary does not preserve all interior distinctions.